EV Market Performance and Insights

Global Market Dynamics, Battery Technology Life-Cycle, and Sustainability Analytics

STUDENT / AUTHOR	PROJECT ID	SUBMISSION DATE	MAXIMUM EVALUATION
Naveen Patidar	Individual Project	27 October 2026	5 Marks

Executive Summary & Industry Background

The global electric vehicle (EV) sector is undergoing exponential growth driven by stringent net-zero emission mandates, falling battery cell costs, and escalating demand for sustainable mobility. This research report presents an integrated business intelligence assessment using Tableau desktop analytics. The primary goal is to examine market share across passenger and commercial vehicle categories, contrast advanced cell chemistry performance (LFP, NMC, NCA, Sodium-Ion, and Solid-State), and evaluate real-world CO_2 mitigation outcomes across major regional manufacturing hubs.

1.55B	$64,921	324.5 km	738K t
TOTAL EV SALES UNITS	AVERAGE VEHICLE PRICE	AVERAGE DRIVING RANGE	ANNUAL CO_2 SAVED

Key

Dashboard Design Principles (Activity 1)

In accordance with the project rubrics, the analytical dashboard was engineered according to eight primary visual analytics principles:

- **Clear and Intuitive Layout:** Cohesive 3-row structured matrix balancing executive KPIs, drill-down distributions, and environmental correlations.
- **Appropriate Visualizations:** Categorical sales represented via horizontal bar charts, geographical production via filled Mapbox spatial views, and chemistry life-cycles via multi-axis scatter plots.
- **Cohesive Theming & Palette:** Uniform color mapping per battery type (LFP, NCA, NMC, Sodium-Ion, Solid-State) and vehicle class across all sheets.
- **Interactive Controls:** Slicers, categorical parameter filters, and cross-worksheet drill-downs for seamless data exploration.

Tableau Interactive Dashboard Overview

The screenshot below displays the primary Tableau analytics interface developed for this project (Sheet tab: *Dashboard 1*), featuring real-time KPI tiles, unit volume breakdowns, regional manufacturing geographic mapping, and battery cycle degradation diagnostics.



Figure 1: Complete Tableau workspace displaying Category Sales, Battery Technology Distribution, Global Manufacturing Map, and CO2 Savings.

Multi-Worksheet Visual Structure & Metric Definition

Worksheet Component	Analytical Method	Primary Finding & Business Insight
Sales by Category	Horizontal bar chart (Annual Sales)	Hatchbacks and Sedans dominate total consumer sales volume, exceeding 75M units each.
S2: Regional Market Share	Pie & donut breakdown	Asia-Pacific accounts for the majority volume, driven by high production capacity.
S2: Battery by Manufacturer	Stacked area chart over vendors	NMC and LFP chemistry constitute over 70% of total manufacturer production streams.
S2: Manufacturing Map	Filled geospatial choropleth	Production concentration is heavily clustered across India, China, Vietnam, and North America.
S3: Battery Tech vs Range	Grouped vertical bar distribution	Luxury Sedans powered by Solid-State and NMC reach highest ranges (550–600 km).
S3: Battery Lifecycle	Scatter plot (Life vs Degradation %)	LFP cells show higher cycle endurance (3000 cycles) with lower degradation rate (<7%).
S3: CO_2 Impact per Class	Bar chart (Tons Saved)	Pickup Trucks and Electric Commercial Fleets yield the highest net CO_2 emissions reduction.

Stakeholder Scenario Analysis & Recommendations

SCENARIO 1 — MARKET STRATEGY

Emily Johnson — EV Market Strategy Director

Strategic Takeaway: High-volume segments (Hatchbacks and Compact Sedans) drive mass consumer adoption, but Luxury Sedans and SUVs yield superior margins and showcase brand capability with 550+ km ranges. Emily should prioritize expanding LFP batteries in mass-market sedans to optimize production cost while preserving high range NMC/Solid-State options for luxury portfolios.

SCENARIO 2 — BUSINESS INTELLIGENCE

David Chen — Automotive Business Intelligence Manager

Benchmarking Takeaway: Geographic mapping confirms manufacturing concentration in the Asia-Pacific corridor (India, China, Vietnam), providing substantial unit-cost advantages. Automotive entrants must establish regional battery pack joint ventures and balance chemistry sourcing between high-durability LFP cells and fast-charging NCA cells to sustain competitive positioning.

SCENARIO 3 — SUSTAINABILITY RESEARCH

Dr. Sarah Williams — Sustainable Mobility Research Analyst

Policy Takeaway: Electric Buses and Commercial Trucks provide the highest annual CO_2 offset (over 45,000 tons per active cluster) and maintain the highest 3-year retained value ($14.5M). Policymakers should target fleet electrification and commercial depot high-power charging subsidies for immediate carbon abatement impact.

Activity 2: Tableau Public Deployment

Published Dashboard Verification Link

The interactive visualization has been verified and published to Tableau Public as required by Activity 2 deliverables:

https://public.tableau.com/views/tab1-17930891578410/Dashboard1?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Student Declaration & Submission Status

I, **Naveen Patidar**, confirm that this analytical report and the accompanying Tableau visualizations accurately represent my individual work on the EV Market Performance and Insights dataset in accordance with SmartBridge evaluation standards.